

HDS06 - Designing and Querying a Clinical Trial Participant Registry

Westbrook University Hospitals NHS Trust, Clinical Trial Participant Registry

Malik J. Lainsbury | [github](#)

2026-07-10



UNIVERSITY OF
CAMBRIDGE

Professional and
Continuing Education

Table of contents

Introduction	3
Part 1: Database Design	4
Entity Relationship Diagram	4
Table Descriptions	5
NULL and NOT NULL Decisions	6
Relationships Between Tables	7
Part 2: SQL	8
INSERT statements	8
SQL Queries	11
Part 3: Reflection	18
Why no participant names?	18
A situation where the data could mislead	18
Why consent records are append-only, and how to enforce it	19
Reference	20
Appendix A — Data dictionary	21
ClinicalTrial	21
HospitalSite	22
Participant	22
InactivationReason	22
TrialSite (link table)	23
ConsentVersion	23
Enrollment	23
ConsentSigning (append-only)	24
Appendix B — File manifest	25
Appendix C — Interoperability mapping (HL7 FHIR)	26
Coversheet	27
MSt Healthcare Data Science — Authentication of practice	27
Details	27
Permission to share your assignment	27
University statement of originality	27
Questions for reflection	28
Appendix D — Additional Operational Data	29

Overview	29
Why a transition, not a second simultaneous enrolment	29
Additional Trial Transition Added	30
Why Ridgeway Community Keeps Zero Enrolments	30
Data Coverage Summary	30
Query 4, 5 and 7 results with this data	30

Introduction

This report designs a fictional relational database for the Westbrook University Hospitals NHS Trust Clinical Trial Participant Registry. It implements it in a Microsoft SQL Server (T-SQL), populates it and answers seven fictional clinical questions. The accompanying runnable script is `script/HDS_DB_LAINSBURY_2026.sql`.

Note

Reproducing the results. Run `script/HDS_DB_LAINSBURY_2026.sql`. It the requested single self-contained script that creates the database, tables, data, trigger and queries in order. It ran on my instance without errors from start to finish. Each query returns one result grid that are screenshotted below. I used the VS Code mssql extension connected to a local SQL Server.

Part 1: Database Design

Entity Relationship Diagram

The DB schema has eight tables. **TrialSite** is necessary addition that resolves the many-to-many relationship between trials and sites. **Enrollment** records a participant's participation in a trial at a site. **ConsentSigning** is the append-only log of consent events to ensure a complete audit trail. The ERD is shown in crow's-foot notation in Figure 1.

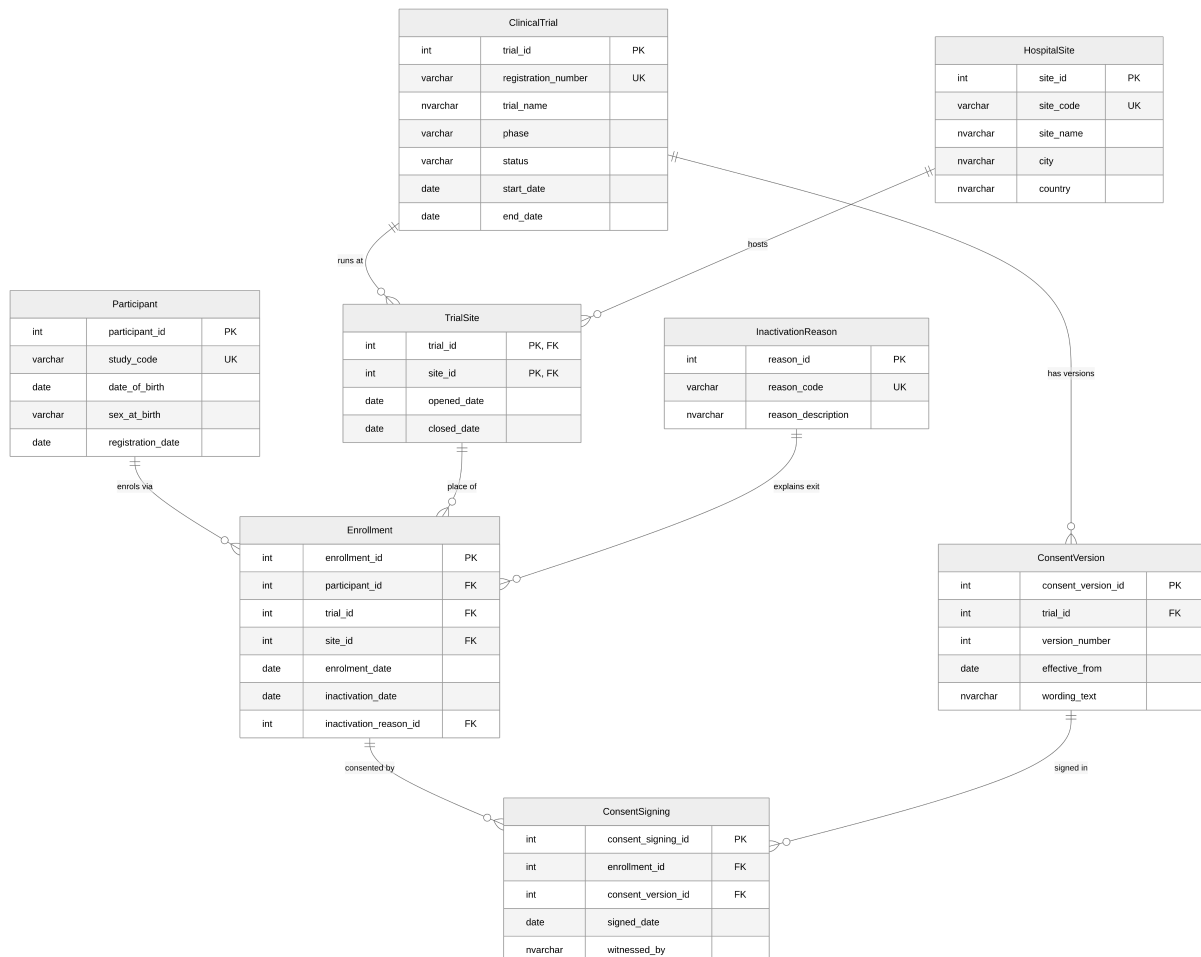


Figure 1: Entity Relationship Diagram (crow's-foot notation).

Table Descriptions

Every table uses a surrogate INT IDENTITY primary key for internal joins, plus one or more **business UNIQUE keys** for the values clinical staff actually recognise. Every foreign key targets the surrogate key, never the business key, so an externally issued code (ISRCTN number, site code, study code) can be corrected in one parent row without cascading through child tables; business keys appear only in the INSERT look-up subqueries.

Table 1: Table descriptions and key design choices

Table	Purpose and key design choices
ClinicalTrial	One row per trial. PK <code>trial_id</code> (IDENTITY). UNIQUE <code>registration_number</code> — the ISRCTN number is the externally recognised, never-reused identifier, so it is the natural key. CHECK on <code>phase</code> (I/II/III/IV) and <code>status</code> enforce the controlled vocabularies; CHECK (<code>end_date</code> >= <code>start_date</code>) prevents impossible ranges. Child tables reference the surrogate <code>trial_id</code> , never the ISRCTN number.
HospitalSite	One row per site. PK <code>site_id</code> . UNIQUE <code>site_code</code> — short codes (e.g. WBK-GEN) are how sites are referred to operationally, so they form the natural key. Child tables reference the surrogate <code>site_id</code> , never <code>site_code</code> .
Participant	One pseudonymised participant per row; no names held . PK <code>participant_id</code> . UNIQUE <code>study_code</code> — the pseudonym is the only safe external identifier. CHECK on <code>sex_at_birth</code> (biological sex at birth, not gender identity — only two values are recorded); CHECK (<code>date_of_birth</code> < <code>registration_date</code>). <code>Enrollment</code> references the surrogate <code>participant_id</code> , never the study code.
InactivationReason	Controlled vocabulary of reasons a participant leaves a trial, so exits report consistently across trials. PK <code>reason_id</code> . UNIQUE <code>reason_code</code> — the stable report token. <code>Enrollment</code> references the surrogate <code>reason_id</code> , so codes can be renamed safely.
TrialSite	Link table resolving the trial–site many-to-many; also records when each site opened/closed for recruitment on that trial. Composite PK (<code>trial_id</code> , <code>site_id</code>) — the pairing itself is the row’s identity, so no surrogate IDENTITY column is used here (unlike every other table); FKs to <code>ClinicalTrial</code> and <code>HospitalSite</code> ; CHECK (<code>closed_date</code> >= <code>opened_date</code>). Both FKs reference the parents’ surrogate keys, not their business codes.
ConsentVersion	Every approved version of a trial’s consent form, including wording and effective date; old versions never overwritten. PK <code>consent_version_id</code> . UNIQUE (<code>trial_id</code> , <code>version_number</code>); FK to <code>ClinicalTrial</code> ; <code>wording_text</code> is NVARCHAR(MAX). The FK references the surrogate <code>trial_id</code> , not the ISRCTN number.

Table	Purpose and key design choices
Enrollment	A participant enrolled in a trial at a specific site. Carries (trial_id , site_id) as a composite FK straight to TrialSite , so trial and site are captured together and a participant can only enrol where the trial actually runs. Exit is recorded without deleting the row. PK enrollment_id ; FKs to Participant , TrialSite (composite), InactivationReason ; UNIQUE (participant_id , trial_id , site_id); paired-null CHECK ; filtered unique index for “one active trial at a time”. All three FKs reference surrogate keys.
ConsentSigning	Append-only log of every consent signing event: when, which version, who witnessed. PK consent_signing_id ; FKs to Enrollment and ConsentVersion ; UNIQUE (enrollment_id , consent_version_id); protected by an append-only trigger. Both FKs reference surrogate keys.

Check constraints.

- Phase/status/sex use **CHECK ... IN (...)** to guarantee clean categorical data for cross-trial reporting.
- Date checks (**end_date** >= **start_date**, **closed_date** >= **opened_date**, **inactivation_date** >= **enrolment_date**) prevent logically impossible records.
- The paired-null check on **Enrollment** guarantees a leaving date never appears without a reason and vice-versa.

NULL and NOT NULL Decisions

Table 2: Nullability decisions and their justification

Column	Decision	Rationale
ClinicalTrial.end_date	NULL	NULL is meaningful: the trial is still running. Current trials have no end date <i>yet</i> .
ClinicalTrial.start_date , status , phase	NOT NULL	A trial cannot exist without these.
TrialSite.closed_date	NULL	The site can still be open for recruitment on that trial.
Enrollment.inactivation_date , inactivation_reason_id	NULL (paired)	The classic “meaningful absence”. The paired-null CHECK keeps them in step.
Participant.date_of_birth , registration_date , sex_at_birth	NOT NULL	Required for eligibility, age and demographic reporting.
ConsentSigning.signed_date , witnessed_by	NOT NULL	A consent event without a date or witness is not a valid consent event.
ConsentVersion.wording_text , effective_from	NOT NULL	A consent version is meaningless without its wording and the date it took effect.

Relationships Between Tables

Table 3: Relationships between tables

Tables	Type / FK	What it represents
ClinicalTrial – HospitalSite	M:N via TrialSite (FK <code>trial_id</code> , <code>site_id</code>)	A trial runs at many sites and a site hosts many trials. The link table also carries the open/close recruitment dates.
ClinicalTrial -> ConsentVersion	1:M (FK <code>trial_id</code>)	One trial accumulates many consent-form versions over its life as the protocol changes.
Participant -> Enrollment	1:M (FK <code>participant_id</code>)	A participant may have several enrolments over time, only one active at once as defined per brief (filtered index).
TrialSite -> Enrollment	1:M (composite FK <code>trial_id</code> , <code>site_id</code>)	Each enrolment happens at exactly one trial-site pairing that the TrialSite link table confirms actually exists.
InactivationReason -> Enrollment	1:M optional (FK <code>inactivation_reason_id</code>)	A standardised reason classifies many exits; active enrolments reference none.
Enrollment -> ConsentSigning	1:M (FK <code>enrollment_id</code>)	One enrolment can have several signing events (original + re-consent after amendment).
ConsentVersion -> ConsentSigning	1:M (FK <code>consent_version_id</code>)	Each signing records which version was signed; one version is signed by many participants.

Many-to-many resolution. The only many-to-many relationship (ClinicalTrial – HospitalSite) is resolved with the TrialSite junction table. I decided to use a composite primary key (`trial_id`, `site_id`). Therefore no surrogate is needed since the pairing itself is the row's identity.

Part 2: SQL

INSERT statements

Every INSERT that needs to reference another table (e.g. Enrollment needs a trial_id) looks up the ID up by a human-recognisable business key instead of hardcoding the raw number. There are no hard-coded IDENTITY integers to avoid mistakes.

Note

I have only shown the first insert per table in this report to illustrate the pattern. All INSERTs are in `script/HDS_DB_LAINSBURY_2026.sql`.

ClinicalTrial

```
INSERT INTO dbo.ClinicalTrial (registration_number, trial_name, phase, status,  
    ↪ start_date, end_date)  
VALUES ('ISRCTN10234567', N'CARDIOPROTECT: Cardiac Rehabilitation in Post-MI  
    ↪ Patients',  
        'III', 'active', '2023-03-01', NULL);
```

HospitalSite

```
INSERT INTO dbo.HospitalSite (site_code, site_name, city, country)  
VALUES ('WBK-GEN', N'Westbrook General Hospital', N'London', N'England');
```

Participant

```
INSERT INTO dbo.Participant (study_code, date_of_birth, sex_at_birth,  
    ↪ registration_date)  
VALUES ('WBK-001', '1958-04-12', 'Male', '2021-11-03');
```

InactivationReason

```
INSERT INTO dbo.InactivationReason (reason_code, reason_description)
VALUES ('consent_withdrawn', N'Participant withdrew their consent to take part
↳ in the trial');
```

TrialSite (FK look-ups via subquery)

```
INSERT INTO dbo.TrialSite (trial_id, site_id, opened_date, closed_date)
VALUES (
  (SELECT trial_id FROM dbo.ClinicalTrial WHERE
↳ registration_number='ISRCTN10234567'),
  (SELECT site_id FROM dbo.HospitalSite WHERE site_code='WBK-GEN'),
  '2023-03-01', NULL);
```

ConsentVersion

```
INSERT INTO dbo.ConsentVersion (trial_id, version_number, effective_from,
↳ wording_text)
VALUES (
  (SELECT trial_id FROM dbo.ClinicalTrial WHERE
↳ registration_number='ISRCTN10234567'),
  1, '2023-03-01',
  N'I agree to take part in the CARDIOPROTECT study. ... I may withdraw at any
↳ time without affecting my medical care.');
```

Enrollment (resolves trial_id and site_id directly from business keys)

```
INSERT INTO dbo.Enrollment (participant_id, trial_id, site_id, enrolment_date,
↳ inactivation_date, inactivation_reason_id)
VALUES (
  (SELECT participant_id FROM dbo.Participant WHERE study_code='WBK-001'),
  (SELECT trial_id FROM dbo.ClinicalTrial WHERE
↳ registration_number='ISRCTN10234567'),
  (SELECT site_id FROM dbo.HospitalSite WHERE site_code='WBK-GEN'),
  '2023-03-15', NULL, NULL);
```

ConsentSigning (resolves enrolment + version by business keys)

```
INSERT INTO dbo.ConsentSigning (enrollment_id, consent_version_id, signed_date,
    witnessed_by)
VALUES (
    (SELECT e.enrollment_id FROM dbo.Enrollment e
        WHERE e.participant_id = (SELECT participant_id FROM dbo.Participant
            WHERE study_code = 'WBK-001')
        AND e.trial_id = (SELECT trial_id FROM dbo.ClinicalTrial
            WHERE registration_number = 'ISRCTN10234567')),
    (SELECT cv.consent_version_id FROM dbo.ConsentVersion cv
        WHERE cv.trial_id = (SELECT trial_id FROM dbo.ClinicalTrial
            WHERE registration_number = 'ISRCTN10234567')
        AND cv.version_number = 1),
    '2023-03-15', N'Nurse A. Okafor');
```

SQL Queries

Query 1: Current consent wording for CARDIOPROTECT

Clinical question. Before a consent discussion, a research nurse needs the current (most recently effective) consent wording for CARDIOPROTECT: version number, effective date, and full wording.

```
SELECT TOP (1) cv.version_number, cv.effective_from, cv.wording_text
FROM    dbo.ConsentVersion cv
JOIN    dbo.ClinicalTrial ct ON ct.trial_id = cv.trial_id
WHERE   ct.trial_name LIKE 'CARDIOPROTECT%'
ORDER  BY cv.effective_from DESC;
```

Screenshot.

```

605 USE HDS_ClinicalTrial;
606 GO
607
608 /* -----
609     QUERY 1
610     Current (most recently effective) consent wording for the CARDIOPROTECT trial.
611     ----- */
612 SELECT TOP (1)
613     cv.version_number,
614     cv.effective_from,
615     cv.wording_text
616 FROM     dbo.ConsentVersion cv
617 JOIN     dbo.ClinicalTrial ct ON ct.trial_id = cv.trial_id
618 WHERE    ct.trial_name LIKE 'CARDIOPROTECT%'
619 ORDER BY cv.effective_from DESC;
620 GO
621

```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

QUERY RESULTS

Results (1)

Messages

	version_number	effective_from	wording_text
1	2	2024-01-15	I agree to take part in the CARDIOPROTECT stu...

Figure 2: Query 1 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. The query filters to CARDIOPROTECT, orders versions by `effective_from` descending and keeps the top row, so it always returns the latest version even after future amendments. *Data quality:* it relies on `effective_from` being accurate, a new version entered with a wrong or missing effective date could show the nurse out-of-date wording.

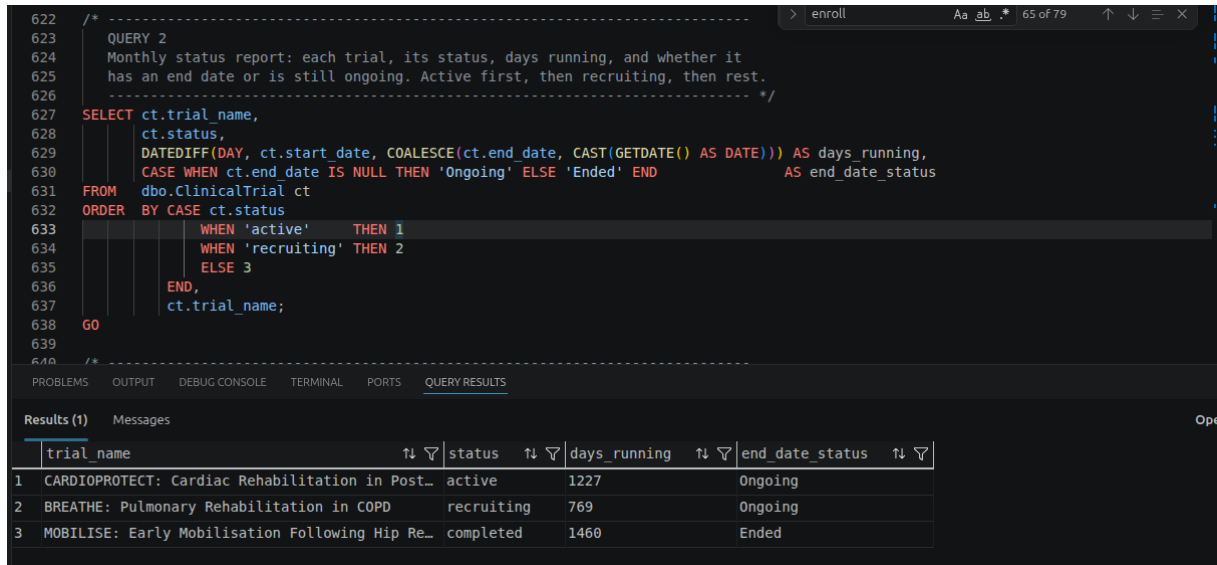
Query 2: Monthly trial status report

Clinical question. A summary of each trial, its status, how long it has run in days, and whether it has an end date or is ongoing; active trials first, then recruiting, then the rest.

```
SELECT ct.trial_name, ct.status,
       DATEDIFF(DAY, ct.start_date, COALESCE(ct.end_date, CAST(GETDATE() AS
-- DATE))) AS days_running,
       CASE WHEN ct.end_date IS NULL THEN 'Ongoing' ELSE 'Ended' END AS
-- end_date_status
FROM    dbo.ClinicalTrial ct
ORDER  BY CASE ct.status WHEN 'active' THEN 1 WHEN 'recruiting' THEN 2 ELSE 3
-- END.
```

```
ct.trial_name;
```

Screenshot.



```
622 /*
623 QUERY 2
624 Monthly status report: each trial, its status, days running, and whether it
625 has an end date or is still ongoing. Active first, then recruiting, then rest.
626 ----- */
627 SELECT ct.trial_name,
628        ct.status,
629        DATEDIFF(DAY, ct.start_date, COALESCE(ct.end_date, CAST(GETDATE() AS DATE))) AS days_running,
630        CASE WHEN ct.end_date IS NULL THEN 'Ongoing' ELSE 'Ended' END AS end_date_status
631 FROM    dbo.ClinicalTrial ct
632 ORDER BY CASE ct.status
633            WHEN 'active' THEN 1
634            WHEN 'recruiting' THEN 2
635            ELSE 3
636        END,
637        ct.trial_name;
638 GO
639 /*
```

trial_name	status	days_running	end_date_status
1 CARDIOPROTECT: Cardiac Rehabilitation in Post...	active	1227	Ongoing
2 BREATHE: Pulmonary Rehabilitation in COPD	recruiting	769	Ongoing
3 MOBILISE: Early Mobilisation Following Hip Re...	completed	1460	Ended

Figure 3: Query 2 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. COALESCE substitutes today’s date for trials with no end date so “days running” is meaningful for ongoing trials, and a CASE expression in ORDER BY gives the custom status priority. *Data quality:* “days running” for ongoing trials changes every day the report is run, so it is only a dated snapshot.

Query 3: Trial history for participant WBK-003

Clinical question. Full history for WBK-003: every trial, enrolment date, inactivation date, and reason code; ordered by enrolment date. Active enrolments have no reason.

```
SELECT ct.trial_name, e.enrolment_date, e.inactivation_date, ir.reason_code
FROM    dbo.Enrollment e
JOIN    dbo.Participant p ON p.participant_id = e.participant_id
JOIN    dbo.ClinicalTrial ct ON ct.trial_id = e.trial_id
LEFT JOIN dbo.InactivationReason ir ON ir.reason_id =
    e.inactivation_reason_id
WHERE   p.study_code = 'WBK-003'
ORDER BY e.enrolment_date;
```

Screenshot.

640 / * -----
641 QUERY 3
642 Full trial history for participant WBK-003. A LEFT JOIN to InactivationReason
643 keeps active enrolments (which have no reason) in the result set.
644 Enrolment stores trial_id directly (composite FK to TrialSite), so
645 ClinicalTrial can be joined straight off it - no TrialSite join needed.
646 ----- */

```

647 SELECT ct.trial_name,
648        e.enrolment_date,
649        e.inactivation_date,
650        ir.reason_code
651 FROM   dbo.Enrolment      e
652 JOIN   dbo.Participant    p ON p.participant_id = e.participant_id
653 JOIN   dbo.ClinicalTrial  ct ON ct.trial_id      = e.trial_id
654 LEFT JOIN dbo.InactivationReason ir ON ir.reason_id = e.inactivation_reason_id
655 WHERE  p.study_code = 'WBK-003'
656 ORDER BY e.enrolment_date;
657 GO
658

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Results (1) Messages

	trial_name	enrolment_date	inactivation_date	reason_code
1	BREATHE: Pulmonary Rehabilitation in COPD	2024-06-10	2024-09-15	consent_withdrawn
2	CARDIOPROTECT: Cardiac Rehabilitation in Post...	2024-10-01	NULL	NULL

Figure 4: Query 3 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. A LEFT JOIN to InactivationReason is essential: an active enrolment has a NULL reason and an INNER join would silently drop it. Enrolment stores trial_id directly (as part of its composite FK to TrialSite), so ClinicalTrial is joined straight off it with no TrialSite hop needed.

Data quality: a participant who left but whose inactivation_reason_id was never populated would show a leaving date with a blank reason, the paired-null check constraint I entered in the table prevents this.

Query 4: Audit of currently active enrolments

Clinical question. Each currently **active** participant with their trial, site, and enrolment date; ordered by trial name then enrolment date.

```

SELECT p.study_code, ct.trial_name, hs.site_name, e.enrolment_date
FROM   dbo.Enrolment      e
JOIN   dbo.Participant    p ON p.participant_id = e.participant_id
JOIN   dbo.ClinicalTrial  ct ON ct.trial_id      = e.trial_id
JOIN   dbo.HospitalSite   hs ON hs.site_id      = e.site_id
WHERE  e.inactivation_date IS NULL
ORDER BY ct.trial_name, e.enrolment_date;

```

Screenshot.

```

659  /* -----
660  QUERY 4
661  Audit of currently ACTIVE enrolments: participant, trial, site, enrol date.
662  Order by trial name then enrolment date.
663  Enrolment stores trial id and site id directly, so ClinicalTrial and
664  HospitalSite are joined straight off it - no TrialSite join needed.
665  ----- */
666  SELECT p.study_code,
667         ct.trial_name,
668         hs.site_name,
669         e.enrolment_date
670  FROM    dbo.Enrollment e
671  JOIN    dbo.Participant p ON p.participant_id = e.participant_id
672  JOIN    dbo.ClinicalTrial ct ON ct.trial_id = e.trial_id
673  JOIN    dbo.HospitalSite hs ON hs.site_id = e.site_id
674  WHERE   e.inactivation_date IS NULL
675  ORDER  BY ct.trial_name, e.enrolment_date;
676  GO

```

	study_code	trial_name	site_name	enrolment_date
1	WBK-001	CARDIOPROTECT: Cardiac Rehabilitation in Post...	Westbrook General Hospital	2023-03-15
2	WBK-003	CARDIOPROTECT: Cardiac Rehabilitation in Post...	Westbrook General Hospital	2024-10-01
3	WBK-002	CARDIOPROTECT: Cardiac Rehabilitation in Post...	St Katherine's Hospital	2025-01-10
4	WBK-005	CARDIOPROTECT: Cardiac Rehabilitation in Post...	St Katherine's Hospital	2025-04-01

Figure 5: Query 4 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. “Active” is expressed as `inactivation_date IS NULL`. `Enrollment` stores `trial_id` and `site_id` directly, so `ClinicalTrial` and `HospitalSite` are joined straight off it, no `TrialSite` hop needed to reconstruct where each active participant sits.

Data quality: the definition of “active” depends on inactivation being recorded promptly. A participant who has left but whose exit was not entered would wrongly appear here.

Query 5: Enrolment count per site (including empty sites)

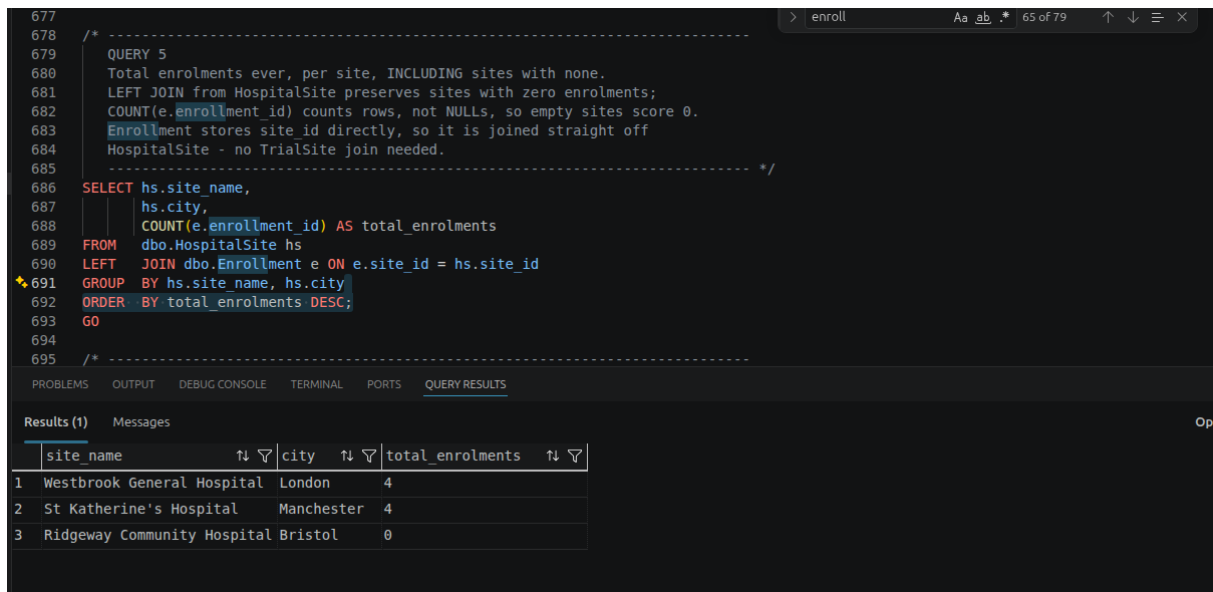
Clinical question. How many participants have ever been enrolled at each site, including sites with none. Show site name, city, total count, descending.

```

SELECT hs.site_name, hs.city, COUNT(e.enrollment_id) AS total_enrolments
FROM    dbo.HospitalSite hs
LEFT    JOIN dbo.Enrollment e ON e.site_id = hs.site_id
GROUP   BY hs.site_name, hs.city
ORDER   BY total_enrolments DESC;

```

Screenshot.



```

677
678
679  /* -----
680  QUERY 5
681  Total enrolments ever, per site, INCLUDING sites with none.
682  LEFT JOIN from HospitalSite preserves sites with zero enrolments;
683  COUNT(e.enrollment_id) counts rows, not NULLs, so empty sites score 0.
684  Enrollment stores site_id directly, so it is joined straight off
685  HospitalSite - no TrialSite join needed.
686  ----- */
687
688  SELECT hs.site_name,
689         hs.city,
690         COUNT(e.enrollment_id) AS total_enrolments
691  FROM   dbo.HospitalSite hs
692  LEFT JOIN dbo.Enrollment e ON e.site_id = hs.site_id
693  GROUP BY hs.site_name, hs.city
694  ORDER BY total_enrolments DESC;
695  GO
696  /* -----

```

site_name	city	total_enrolments
1 Westbrook General Hospital	London	4
2 St Katherine's Hospital	Manchester	4
3 Ridgeway Community Hospital	Bristol	0

Figure 6: Query 5 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. Starting from `HospitalSite` with a `LEFT JOIN` preserves sites with no enrolments, and `COUNT(e.enrollment_id)` counts rows rather than `NULLs` so empty sites correctly score 0. `Enrollment` stores `site_id` directly, so it is joined straight off `HospitalSite` with no `TrialSite` hop needed.

Data quality: A site that participates in a trial but has not yet recorded any enrolments is indistinguishable from one that genuinely has none, the zero needs interpretation in context.

Query 6: Trials with an amended protocol

Clinical question. Trials where the protocol was amended (more than one consent version): trial name, current status, number of versions; descending; only where more than one version exists.

```

SELECT ct.trial_name, ct.status, COUNT(cv.consent_version_id) AS version_count
FROM   dbo.ClinicalTrial ct
JOIN   dbo.ConsentVersion cv ON cv.trial_id = ct.trial_id
GROUP BY ct.trial_name, ct.status
HAVING COUNT(cv.consent_version_id) > 1
ORDER BY version_count DESC;

```

Screenshot.


```

694
695 /* -----
696 QUERY 6
697 Trials with an amended protocol (more than one consent version issued).
698 ----- */
699 SELECT ct.trial_name,
700        ct.status,
701        COUNT(cv.consent_version_id) AS version_count
702 FROM    dbo.ClinicalTrial ct
703 JOIN    dbo.ConsentVersion cv ON cv.trial_id = ct.trial_id
704 GROUP BY ct.trial_name, ct.status
705 HAVING COUNT(cv.consent_version_id) > 1
706 ORDER BY version_count DESC;
707 GO
708
709 /* -----
710 QUERY 7
711 Most recently enrolled participant at each site (across all trials)
712 ----- */

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Results (1) Messages

	trial_name	status	version_count
1	CARDIOPROTECT: Cardiac Rehabilitation in Post...	active	2

Figure 7: Query 6 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. Grouping by trial and counting versions, then filtering with `HAVING COUNT(...) > 1`, isolates amended trials (`HAVING` filters the aggregate, which `WHERE` cannot).

Data quality: this counts version *rows*, so a duplicate version entered in error would overstate the number of genuine amendments.

Query 7: Most recently enrolled participant per site

Clinical question. The most recently enrolled participant at each site across all trials: study code, trial name, site name, enrolment date.

```

WITH RankedEnrolments AS (
    SELECT p.study_code, ct.trial_name, hs.site_name, e.enrolment_date,
           RANK() OVER (PARTITION BY hs.site_id
                        ORDER BY e.enrolment_date DESC) AS rn
    FROM    dbo.Enrollment e
    JOIN    dbo.Participant p ON p.participant_id = e.participant_id
    JOIN    dbo.ClinicalTrial ct ON ct.trial_id = e.trial_id
    JOIN    dbo.HospitalSite hs ON hs.site_id = e.site_id
)
SELECT study_code, trial_name, site_name, enrolment_date
FROM    RankedEnrolments
WHERE   rn = 1
ORDER BY site_name;

```

Screenshot.

```

716
717 WITH RankedEnrolments AS (
718     SELECT p.study_code,
719            ct.trial_name,
720            hs.site_name,
721            e.enrolment_date,
722            RANK() OVER (PARTITION BY hs.site_id
723                        ORDER BY e.enrolment_date DESC) AS rn
724     FROM   dbo.Enrollment e
725     JOIN   dbo.Participant p ON p.participant_id = e.participant_id
726     JOIN   dbo.ClinicalTrial ct ON ct.trial_id = e.trial_id
727     JOIN   dbo.HospitalSite hs ON hs.site_id = e.site_id
728 )
729 SELECT study_code,
730        trial_name,
731        site_name,
732        enrolment_date
733 FROM   RankedEnrolments
734 WHERE  rn = 1
735 ORDER BY site_name;
736 GO
737
738

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Results (1) Messages

	study_code	trial_name	site_name	enrolment_date
1	WBK-005	CARDIOPROTECT: Cardiac Rehabilitation in Post...	St Katherine's Hospital	2025-04-01
2	WBK-003	CARDIOPROTECT: Cardiac Rehabilitation in Post...	Westbrook General Hospital	2024-10-01

Figure 8: Query 7 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. `RANK()` partitioned by site and ordered by enrolment date descending ranks each site’s enrolments; keeping `rn = 1` returns the newest per site. Because tied dates share a rank, two participants enrolled at the same site on the same day would both be returned rather than one being picked arbitrarily. `Enrollment` stores `trial_id` and `site_id` directly, so `ClinicalTrial` and `HospitalSite` are joined straight off it with no `TrialSite` hop needed.

Data quality: enrolment is recorded only to the day, so joint “most recent” participants are indistinguishable; a finer timestamp would be needed to single one out.

Part 3: Reflection

Why no participant names?

Storing only a `study_code` pseudonymises the registry. This follows the data-minimisation principle from the GDPR *Principles relating to processing of personal data* (GDPR Article 5(1)(c))¹. Additional, we could also refer to the Caldicott Principles² of justifying every use of confidential information and using the minimum necessary which reinforces the article 5 of the GDPR. This means a breach of the trial database would not directly expose personal identities and personal identities does not bring necessary information to the database to justify their inclusion.

Direct identifiers such as names would live in a separate, access-controlled master linkage list held by the research office. Researchers can query and report entirely on `study_code`, so trial analyses can be shared more freely because they carry no personal identifiers. The cost is friction and a dependency on the linkage to the access-controlled master linkage list: a broken or out-of-date mapping could mean a participant cannot be reached for a safety issue. The design therefore trades day-to-day convenience for a much stronger confidentiality and information-governance position.

A situation where the data could mislead

Consider Query 4 and Query 5, which both depend on `inactivation_date`. If a participant has actually left a trial but their exit has not yet been entered, they will still appear as “active” and will be counted in active-enrolment reports — overstating recruitment and potentially prompting unnecessary follow-up contact. The data is not *wrong* field by field; it is *stale*, which is harder to spot. A data manager could detect this by reconciling the registry against source records: cross-checking active enrolments against clinic attendance or against the standardised `InactivationReason` log, and flagging participants with no recent activity (e.g. no consent or visit events for an unusually long period) for review. Preventively, they could add a periodic data-quality report listing long-inactive “active” enrolments, set a service standard for how quickly exits must be recorded, and use the `lost_to_follow_up` reason code consistently so genuine non-contact is captured rather than left as a silent gap. The fix is process plus monitoring, not just a query.

Why consent records are append-only, and how to enforce it

Consent records are the legal and ethical evidence that each participant agreed to the specific protocol version in force **at the time**. They must be append-only because consent is a historical fact. This requires trial-critical records to carry a complete, tamper-evident audit trail, and editing or deleting a signing event would destroy the evidence that regulators, sponsors and ethics committees rely on to prove valid consent existed. If the rule were broken, a re-consent after a protocol amendment could be silently overwritten or a withdrawn consent erased, making it impossible to demonstrate which wording a participant actually agreed to and undermining every downstream analysis that assumes valid consent.

I enforced this at the database level with an **INSTEAD OF UPDATE, DELETE trigger** on `ConsentSigning` that rejects any such operation with an error, while leaving `INSERT` free (see `script/HDS_DB_LAINSBURY_2026.sql`). This guarantees the rule regardless of which application or user connects, which a permission grant alone would not.

```
CREATE OR ALTER TRIGGER dbo.trg_ConsentSigning_AppendOnly
ON dbo.ConsentSigning
INSTEAD OF UPDATE, DELETE
AS
BEGIN
    SET NOCOUNT ON;
    THROW 51000, 'ConsentSigning is append-only: rows cannot be updated or
    ↵ deleted.', 1;
END;
```

Reference

Appendix A — Data dictionary

A complete column-level reference for every table, derived from the `CREATE TABLE` section of `script/HDS_DB_LAINSBURY_2026.sql`. *Key*: PK = primary key, FK = foreign key, UK = unique (business) key. All surrogate `*_id` keys are `INT IDENTITY(1,1)`.

This dictionary is not static documentation: the metadata section of `script/HDS_DB_LAINSBURY_2026.sql` stores every table description — and the column descriptions where meaning is not self-evident, in particular the meaningful-NULL columns — **in the database catalog itself** as `MS_Description` extended properties. The dictionary can therefore be regenerated at any time (or consumed by data-catalogue tooling) by querying the system views, so the metadata travels with the schema rather than living in a document that can drift out of date:

```
SELECT t.name AS table_name,
       c.name AS column_name,
       TYPE_NAME(c.user_type_id) AS data_type,
       IIF(c.is_nullable = 1, 'NULL', 'NOT NULL') AS nullability,
       CAST(ep.value AS NVARCHAR(400)) AS column_description
FROM sys.tables t
JOIN sys.columns c ON c.object_id = t.object_id
LEFT JOIN sys.extended_properties ep
      ON ep.major_id = c.object_id AND ep.minor_id = c.column_id
      AND ep.class = 1 AND ep.name = 'MS_Description'
ORDER BY t.name, c.column_id;
```

ClinicalTrial

Table 4: ClinicalTrial

Column	Type	Null	Key	Notes
<code>trial_id</code>	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
<code>registration_number</code>	VARCHAR(20)	NOT NULL	UK	ISRCTN number; externally recognised natural key.
<code>trial_name</code>	NVAR- CHAR(150)	NOT NULL		Full trial title.
<code>phase</code>	VARCHAR(4)	NOT NULL		CHECK IN ('I','II','III','IV').
<code>status</code>	VARCHAR(20)	NOT NULL		CHECK IN ('recruiting','active','completed','suspended').

Column	Type	Null	Key	Notes
<code>start_date</code>	DATE	NOT NULL		Trial start date.
<code>end_date</code>	DATE	NULL		NULL = ongoing. CHECK (<code>end_date</code> >= <code>start_date</code>).

HospitalSite

Table 5: HospitalSite

Column	Type	Null	Key	Notes
<code>site_id</code>	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
<code>site_code</code>	VARCHAR(10)	NOT NULL	UK	Short operational code, e.g. WBK-GEN.
<code>site_name</code>	NVAR- CHAR(100)	NOT NULL		Hospital site name.
<code>city</code>	NVAR- CHAR(60)	NOT NULL		City.
<code>country</code>	NVAR- CHAR(60)	NOT NULL		Country.

Participant

Table 6: Participant

Column	Type	Null	Key	Notes
<code>participant_id</code>	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
<code>study_code</code>	VAR- CHAR(15)	NOT NULL	UK	Pseudonym; only external identifier (no names held).
<code>date_of_birth</code>	DATE	NOT NULL		CHECK (<code>date_of_birth</code> < <code>registration_date</code>).
<code>sex_at_birth</code>	VAR- CHAR(10)	NOT NULL		CHECK IN ('Male','Female'). Records biological sex at birth, not gender identity — only two values are captured for this field.
<code>registration_date</code>	DATE	NOT NULL		Date first registered with the research office.

InactivationReason

Table 7: InactivationReason

Column	Type	Null	Key	Notes
<code>reason_id</code>	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
<code>reason_code</code>	VARCHAR(30)	NOT NULL	UK	Standardised report token, e.g. <code>consent_withdrawn</code> .

Column	Type	Null	Key	Notes
reason_description	NVAR-CHAR(255)	NOT NULL		Human-readable description of the reason.

TrialSite (link table)

Table 8: TrialSite. The only table in the schema without a surrogate IDENTITY key — the (trial_id, site_id) pairing is itself the row’s identity.

Column	Type	Null	Key	Notes
trial_id	INT	NOT NULL	PK, FK	-> ClinicalTrial . Part of the composite PK (trial_id, site_id).
site_id	INT	NOT NULL	PK, FK	-> HospitalSite . Part of the composite PK (trial_id, site_id).
opened_date	DATE	NOT NULL		Date the site opened for recruitment on this trial.
closed_date	DATE	NULL		NULL = still open. CHECK (closed_date >= opened_date).

ConsentVersion

Table 9: ConsentVersion

Column	Type	Null	Key	Notes
consent_version_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
trial_id	INT	NOT NULL	FK, UK	-> ClinicalTrial . Part of UNIQUE (trial_id, version_number).
version_number	INT	NOT NULL	UK	CHECK (version_number > 0). Part of the unique key.
effective_from	DATE	NOT NULL		Date this version came into effect.
wording_text	NVAR-CHAR(MAX)	NOT NULL		Full consent-form wording; never overwritten.

Enrollment

Table 10: Enrollment. A filtered unique index allows only one active enrolment per participant (WHERE inactivation_date IS NULL).

Column	Type	Null	Key	Notes
enrollment_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
participant_id	INT	NOT NULL	FK, UK	-> Participant . Part of UNIQUE (participant_id, trial_id, site_id).
trial_id	INT	NOT NULL	FK, UK	Composite FK with site_id -> TrialSite (confirms the site actually runs the trial). Part of the unique key.

Column	Type	Null	Key	Notes
site_id	INT	NOT NULL	FK, UK	Composite FK with trial_id -> TrialSite. Part of the unique key.
enrolment_date	DATE	NOT NULL		Date enrolment began.
inactivation_date	DATE	NULL		NULL = still active. CHECK (inactivation_date >= enrolment_date).
inactivation_reason_id	INT	NULL	FK	-> InactivationReason. Paired-null CHECK with inactivation_date.

ConsentSigning (append-only)

Table 11: ConsentSigning. Protected by an INSTEAD OF UPDATE, DELETE trigger so rows can only ever be inserted.

Column	Type	Null	Key	Notes
consent_signing_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
enrollment_id	INT	NOT NULL	FK, UK	-> Enrollment. Part of UNIQUE (enrollment_id, consent_version_id).
consent_version_id	INT	NOT NULL	FK, UK	-> ConsentVersion. Part of the unique key.
signed_date	DATE	NOT NULL		Date the consent was signed.
witnessed_by	NVAR- CHAR(100)	NOT NULL		Name/role of the witnessing staff member.

Appendix B — File manifest

The submission is a single, self-contained script, run top to bottom:

Table 12: `script/HDS_DB_LAINSBURY_2026.sql` — section contents in run order

Section (in run order)	Contents
Create database	(Re)creates the <code>HDS_ClinicalTrial</code> database
Create tables	<code>CREATE TABLEs</code> , constraints, indexes
Insert reference data	The five reference sheets from the brief
Insert additional operational data	<code>TrialSite</code> , <code>Enrollment</code> , <code>ConsentSigning</code> data
Append-only trigger	<code>INSTEAD OF UPDATE</code> , <code>DELETE</code> enforcement on <code>ConsentSigning</code>
Queries	The seven queries
Metadata	Extended-property metadata + catalog-driven data dictionary

Appendix C — Interoperability mapping (HL7 FHIR)

The registry does not exist in isolation: if the trust later needs to exchange trial data with other NHS organisations, sponsors, or national research platforms, the natural target is the HL7 FHIR standard used across the NHS. The schema maps cleanly onto FHIR resources, which validates the entity choices and gives a ready-made route for future data linkage beyond the trust:

Table 13: Mapping of registry tables to HL7 FHIR resources

Registry table	FHIR resource	Notes
ClinicalTrial	ResearchStudy	Phase and status map to the resource's phase and status elements.
Participant	Patient (pseudonymised)	study_code becomes a pseudonymous identifier ; no name elements populated.
Enrollment	ResearchSubject	FHIR likewise links one patient to one study with period and status.
HospitalSite	Location / Organization	Site code as the identifier .
ConsentSigning + ConsentVersion	Consent	FHIR Consent also references the specific policy/form version agreed to.
InactivationReason	ResearchSubject progress reason (coded)	The controlled vocabulary would be published as a FHIR CodeSystem .

Coversheet

MSt Healthcare Data Science — Authentication of practice

I confirm that I have fully read and understood the assignment brief for this module. **Y**

Details

Name	Malik J. Lainsbury
Submission Date	03 – 07 – 26
Word Count: whole assignment including codes	~3,000 word-equivalents (1,560 prose + 7 queries x 150 + 8 INSERT statements x 50)
Word Count: main body excluding abstract, references and supplementary materials	1,560 prose words (appendices excluded as supplementary)

Permission to share your assignment

I do give permission to share my assignment with future MSt participants.

University statement of originality

This assignment is the result of my own work and includes nothing which is the outcome of work done in collaboration except as declared in the Preface and specified in the text. It is not substantially the same as any that I have previously submitted for a degree or diploma or other qualification at the University of Cambridge or any other university or similar institution, or that is being concurrently submitted, except as declared in the Preface and specific in the text. I further state that no substantial part of my Portfolio has already been submitted, nor is being concurrently submitted for any such degree, diploma or other qualification at the University of

Cambridge or any other university or similar institution except as declared in the Preface and specified in the text.

I confirm the statement of originality as above **Y**

Questions for reflection

1. Which aspects of this assignment are you most uncertain about and/or would most like to receive feedback on? Whether giving `Enrollment` a composite FK (`trial_id`, `site_id`) straight to `TrialSite` (rather than a single surrogate `trial_site_id` pointer) is considered good practice or over-normalisation; and whether the filtered unique index is an acceptable way to express the “one active trial at a time” rule.

5. Using the wording in the rubric, how would you describe the quality of the different aspects of your work? The ERD, table descriptions and relationships aim at the “high performance” descriptors: all entities identified, the many-to-many resolved with a link table, and each decision justified against the scenario. The script runs end-to-end without errors and every foreign key is resolved by subquery on a business key.

Appendix D — Additional Operational Data

Overview

Assignment section 6.1 specifies: “In addition to the reference data provided, you must also add **your own data** to all the other records you have designed and there should be **enough data of your own to answer all seven queries meaningfully**.”

The reference data provided (ClinicalTrial, HospitalSite, InactivationReason, Participant, ConsentVersion) contains no TrialSite, Enrollment, or ConsentSigning rows at all — those three tables are entirely self-authored operational data. Beyond the minimum needed to satisfy the six worked-example requirements, one further **trial transition** was added to enrich Queries 4 and 7, and Ridgeway Community is deliberately left with **zero enrolments** so that Query 5 demonstrably includes a site with none. TrialSite uses a composite primary key (trial_id, site_id) rather than a surrogate IDENTITY (see Table Descriptions, main body); as a result Enrollment stores trial_id and site_id directly rather than a single trial_site_id pointer.

Why a transition, not a second simultaneous enrolment

The schema enforces the brief’s rule that a participant may be actively enrolled in only one trial at a time with a filtered unique index:

```
CREATE UNIQUE INDEX UX_Enrollment_OneActivePerParticipant
ON dbo.Enrollment(participant_id)
WHERE inactivation_date IS NULL;
```

This permits at most one row per participant with inactivation_date IS NULL. Consequently, the addition below **closes** the participant’s existing active enrolment first (an UPDATE in place, preserving the original row as the brief requires), then **inserts** a new active enrolment — exactly the “left a previous trial and later joined a new one” pattern the scenario itself describes.

Additional Trial Transition Added

WBK-005: BREATHE (Westbrook General) closed, CARDIOPROTECT (St Katherine's) opened

Data added: - **UPDATE:** WBK-005's active BREATHE enrolment at Westbrook General is closed — `inactivation_date = 2025-03-15, inactivation_reason_id = clinician_decision` - **INSERT:** new active enrolment — CARDIOPROTECT at St Katherine's, `enrolment_date = 2025-04-01` (the most recent enrolment date in the whole database) - **INSERT:** consent signing — CARDIOPROTECT v2 (the current version), signed 2025-04-01, witnessed by Nurse A. Okafor

Impact on queries: - **Query 7:** gives St Katherine's a genuine choice between two dated candidates (WBK-002's original 2025-01-10 enrolment and WBK-005's 2025-04-01 enrolment), so the `RANK() OVER (PARTITION BY site ...)` logic is doing real selection work rather than returning a trivial single-row partition - **Query 4:** WBK-005 still appears (now under CARDIOPROTECT/St Katherine's instead of BREATHE/Westbrook General)

Why Ridgeway Community Keeps Zero Enrolments

Ridgeway Community is linked to BREATHE in `TrialSite` — it participates in the trial — but no participant is ever enrolled there. This is deliberate: Query 5 asks for enrolment counts *including sites that currently have no enrolments recorded*, and only a genuinely empty site proves the `LEFT JOIN` preserves the zero row (an `INNER JOIN` would silently drop it). It also reflects a real registry situation: a site can be open for recruitment before its first participant arrives.

Data Coverage Summary

Metric	Before addition	After addition
Enrollment rows	7	8
Active enrolments	4	4 (unchanged — the transition closes one, opens one)
Hospital sites with \$ \$1 enrolment ever	2	2 (Ridgeway deliberately kept at zero for Query 5)
Consent signing events	8	9
Trials each active participant has moved between	0	1 participant (WBK-005) demonstrates the “left one trial, joined another” pattern

Query 4, 5 and 7 results with this data

Query 4 (active enrolments, 4 rows): WBK-001 / CARDIOPROTECT / Westbrook General / 2023-03-15; WBK-003 / CARDIOPROTECT / Westbrook General / 2024-10-01; WBK-002 / CARDIOPROTECT / St Katherine's / 2025-01-10; WBK-005 / CARDIOPROTECT / St Katherine's / 2025-04-01.

Query 5 (enrolments per site, 3 rows): Westbrook General 4; St Katherine's 4; Ridgeway Community 0 — the zero row is preserved by the `LEFT JOIN`.

Query 7 (most recent per site, 2 rows): St Katherine's — WBK-005 / CARDIOPROTECT / 2025-04-01; Westbrook General — WBK-003 / CARDIOPROTECT / 2024-10-01. Ridgeway Community has no enrolments and therefore no row.

1. Art. 5 GDPR – Principles relating to processing of personal data. *General Data Protection Regulation (GDPR)*.
2. The Caldicott Principles. *GOVUK*.